Zhongtian Zhang

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Appointments

2025–2027	**Harry H. Hess Postdoctoral Fellow**, Department of Geosciences, Princeton University (starting September 2025)
2023–2025	**Carnegie Postdoctoral Fellow**, Earth and Planets Laboratory, Carnegie Institution for Science

Education

2023	**Ph.D.**, Earth & Planetary Sciences, Yale University
2019	**M.Phil.**, Geology & Geophysics, Yale University
2017	**B.Sc.**, Geology, Nanjing University

Publications

Articles: Submitted

[15] Grewal, D., **Zhang, Z.**, Manilal, V., Kruijer, T., Bottke, W., Stewart, S., Protracted core formation and impact disruptions shaped the earliest outer solar system planetesimals. Under revision.

Articles: Published or In-Press

[14] **Zhang, Z.**, Luo, H., Hao, M., Deng, J., Regimes of element transfer between Earth's core and basal magma ocean. J. Geophys. Res., Solid Earth. Accepted.

[13] **Zhang, Z.**, Driscoll, P. E., 2025. Inefficient loss of moderately volatile elements from exposed planetesimal magma oceans. J. Geophys. Res., Planets 130 (2), e2024JE008671. https://doi.org/10.1029/2024JE008671

[12] **Zhang, Z.**, Wang, J., 2024. Quantification of classical and non-classical crystallization pathways in calcite precipitation. Earth Planet. Sci. Lett. 636, 118712. https://doi.org/10.1016/j.epsl.2024.118712.

[11] **Zhang, Z.**, 2024. Trace elements in IVA iron meteorites explained by limited solid-liquid equilibration during inward solidification. Icarus. 408, 115860. https://doi.org/10.1016/j.icarus.2023.115860.

[10] **Zhang, Z.**, 2023. Ice sublimation in planetesimals formed at the outward migrating snowline. Astrophys. J. Lett. 956 (1), L25. https://doi.org/10.3847/2041-8213/acfdaa

[9] **Zhang, Z.**, Bercovici, D., 2023. Generation of a measurable magnetic field in a metal asteroid with a rubble-pile inner core. Proc. Natl. Acad. Sci. 120 (32), e2221696120. https://doi.org/10.1073/pnas.2221696120 (Press releases: Yale News, APS Physics Magazine)

[8] Gong, Z., Evans, D. A. D., **Zhang, Z.**, Yan, C., 2023. Mid-Proterozoic geomagnetic field was more consistent with a dipole than a quadrupole. Geology 51 (6), 571–575. https://doi.org/10.1130/G50941.1

[7] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2023. Melt migration in rubble-pile planetesimals: Implications for the formation of primitive achondrites. Earth Planet. Sci. Lett. 605, 118019. https://doi.org/10.1016/j.epsl.2023.118019

[6] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2022. Cold compaction and macro-porosity removal in rubble-pile asteroids: 2. Applications. J. Geophys. Res., Planets 127 (10), e2022JE007343. https://doi.org/10.1029/2022JE007343

[5] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2022. Cold compaction and macro-porosity removal in rubble-pile asteroids: 1. Theory. J. Geophys. Res., Planets 127 (10), e2022JE007342. https://doi.org/10.1029/2022JE007342

[4] **Zhang, Z.**, Bercovici, D., Jordan, J.S., 2021. A two-phase model for the evolution of planetary embryos with implications for the formation of Mars. J. Geophys. Res., Planets 126 (4), e2020JE006754. https://doi.org/10.1029/2020JE006754 (Editor's highlight)

[3] **Zhang, Z.**, Karato, S.-I., 2021. The effect of pressure on grain-growth kinetics in olivine aggregates with some geophysical applications. J. Geophys. Res., Solid Earth 126 (2), e2020JB020886. https://doi.org/10.1029/2020JB020886

[2] **Zhang, Z.**, Wu, B., Wang, T., Hui, H., 2020. Settling of immiscible droplets: A theoretical model for the missing link between microscopic and outcrop observations. J. Geophys. Res., Solid Earth 125 (6), e2019JB018829. https://doi.org/10.1029/2019JB018829

[1] Xu, Y., Tang, W., Hui, H., Rudnick, R. L., Shang, S., **Zhang, Z.**, 2019. Reconciling the discrepancy between the dehydration rates in mantle olivine and pyroxene during xenolith emplacement. Geochim. Cosmochim. Acta. 267, 179–195. https://doi.org/10.1016/j.gca.2019.09.023

Invited seminars

2025	Department of Earth and Planetary Sciences, Rutgers University
	Department of Geophysical Sciences, The University of Chicago
	Earth and Planets Laboratory, Carnegie Institution for Science
2024	Department of Geosciences, Princeton University
	Department of Mineral Sciences, Smithsonian National Museum of Natural History
	Department of Geosciences, Stony Brook University
	Division of Geological and Planetary Sciences, California Institute of Technology

Honors

2025	**Harry H. Hess Postdoctoral Fellowship** (Department of Geosciences, Princeton University)
2023	**Carnegie Postdoctoral Fellowship** (Earth and Planets Lab, Carnegie Institution for Science)
2022	**Elias Loomis Prize** (Department of Earth & Planetary Sciences, Yale University)

Teaching experience

Teaching Fellowship at Yale University

Fall 2020	EPS 456/556 Introduction to Seismology
Spring 2020	G&G275b/F&ES716b Renewable Energy
Fall 2019	G&G 428/528 Science of Complex Systems

Service

Peer review for Journal of Geophysical Research: Planets (4), Science Advances (1), Geophysical Research Letters (1), Geochimica et Cosmochimica Acta (1), Icarus (1), Physics of the Earth and Planetary Interiors (1), iScience (1)

Proposal panelist for NASA (1)

Proposal reviewer (non-panelist) for NASA (1), European Research Council (1)

Session conveyor for AGU Fall Meeting, DI009: Insights into planetary formation and evolution from interdisciplinary perspectives (2024)

Colloquium committee, Department of Earth & Planetary Sciences, Yale University (2018–2021)

Treasurer, Dana Club, Department of Earth & Planetary Sciences, Yale University (2019–2020)

IC-FG (International Community and First Generation students) working group, IDEA (Inclusion, Diversity, Equity, Anti-racism and Anti-discrimination) committee of Department of Earth & Planetary Sciences, Yale University (2022–2023)

DiMhCi (Disability, Mental health, and Chronic illness) working group, IDEA committee of Department of Earth & Planetary Sciences, Yale University (2022–2023)